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Iteration: ExercisesIntroduction to computer programmingMan

Introduction to Computer Programming

Management and Artificial Intelligence

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Exercises

Exercise 1: Coin flipping

- Ask the user how many coin tosses have to be performed
- Keep track of the number of times we get heads and tails
- We assume a *fair coin* (i.e., unbiased 50% chance of getting head and tail at each toss)

HINT: You may use the `random.randint()` function to simulate a single coin toss.

```
import random

tail = 0
head = 0

n = 3 # number of coin tosses
# n = int(input("How many coin tosses? "))
for i in range(n):
    x = random.randint(1, 100)
    if x > 50:
        head = head + 1
    else:
        tail = tail + 1

print("Number of heads: ", head)
print("Number of tails: ", tail)
```

Number of heads: 1
Number of tails: 2

Exercise 2: Sum of sequence

Write a function named `rolling_sum()` that:

- repeatedly asks the user to insert a number
- we assume integer numbers
- keeps track of the sum of the numbers inserted so far
- if the user inserts 0, then exit and return the sum

```
%%script echo skipping
```

```
def rolling_sum():  
    this_number = int(input("Insert a number: "))  
    my_sum = this_number  
  
    while this_number != 0:  
        this_number = int(input("Insert a number: "))  
        my_sum = my_sum + this_number  
  
    return my_sum  
  
print("Total sum is: ", rolling_sum())
```

skipping

Exercise 3: Odd or even?

Write a function named `odd_or_even()` that:

- accepts an integer number, say `n`
- prints the number of odd numbers in range `[1,n]` (i.e., both included)
- prints the number of even numbers in range `[1,n]` (i.e., both included)

```
def odd_or_even(n):  
    odd = 0  
    even = 0  
  
    for i in range(1, n + 1):  
        if i % 2 == 0:  
            even += 1  
        else:  
            odd += 1  
  
    print("Number of odd elements: ", odd)  
    print("Number of even elements: ", even)  
  
odd_or_even(10)
```

Number of odd elements: 5
Number of even elements: 5

Exercise 4: The Fibonacci Sequence

Write a function named `fibonacci1()` that:

- accepts an integer number, say `n`
- prints the first `n` elements of the Fibonacci sequence.

Definition: Each number in the Fibonacci sequence is given by the sum of the two preceding ones. The Fibonacci sequence starts with two 'seed' numbers `0` and `1` and then the proper sequence starts: `0, 1, 1, 2, 3, 5, 8, 13, 21, ...`

```
def fibonacci1(n):  
    second_to_last = 0  
    last = 1  
  
    print(second_to_last)  
    print(last)  
  
    for i in range(n - 2):  
        new = last + second_to_last  
        print(new)  
        second_to_last = last  
        last = new  
  
fibonacci1(10)
```

0
1
1
2
3
5
8
13
21
34

Exercise 5: Multiples

Write a function named `multiples()` that:

- accepts an integer number, say `n`
- iterates the integers in `[1, n]`
- for multiples of 3, print `'Mult3'` instead of the number
- for multiples of 5, print `'Mult5'` instead of the number
- for multiples of 5 and 3, print `'Mult35'` instead of the number

```
def multiples(n):  
    for num in range(1, n + 1):  
        if num % 3 == 0 and num % 5 == 0:  
            print("Mult35")  
        elif num % 3 == 0:  
            print("Mult3")  
        elif num % 5 == 0:  
            print("Mult5")  
        else:  
            print(num)
```

multiples(15)

1
2
Mult3
4
Mult5
Mult3
7
8
Mult3
Mult5
11
Mult3
13
14
Mult35

Exercise 6: Print a subset

Write a function named `print_subset()` that:

- accepts an integer number, say n
- accepts an integer number, say $m \leq n$
- scans all integer numbers in range $[1, n]$, but prints all numbers in range $[1, m]$

Solution 1

```
def print_subset(n, m):  
    for num in range(1, n + 1):  
        if num <= m:  
            print(num)  
  
print_subset(5, 3)
```

1
2
3

Solution 2

```
def print_subset(n, m):  
    for num in range(1, n + 1):  
        print(num)  
        if num == m:  
            break  
  
print_subset(5, 3)
```

1
2
3

Stop and think: Under some circumstances, the second solution is more efficient than the first. What are those circumstances and why is it more efficient?

Exercise 7: Primality test

Write a function named `smallerPrimes()` that:

- accepts a positive integer number, say n
- prints all positive integer prime numbers smaller than n

Recall: A *prime* number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Hint: The simplest primality test aims at checking whether n is evenly divisible (i.e., division with no remainder) by any number in range $[2, n/2]$. If so, the number is not prime.


```
def is_prime(num):  
    if num <= 1:  
        return False  
    for i in range(2, int(num/2) + 1):  
        if num % i == 0:  
            return False  
    return True  
  
def smallerPrimes(n):  
    for i in range(2, n):  
        if is_prime(i):  
            print(i)  
  
smallerPrimes(20)
```

```
2  
3  
5  
7  
11  
13  
17  
19
```

Exercise 8: Pythagorean triples

Write a function named `pythagoreanTriples()` that:

- accepts an integer positive number, say n
- prints all x and y pairs such that $x^2 + y^2 = n^2$

Recall: The Pythagorean triple consists of three positive integers x , y and n , with $x, y \leq n$, such that: $x^2 + y^2 = n^2$

```
def pythagoreanTriples(n):  
    for x in range(1, n):  
        for y in range(1, n):  
            if x**2 + y**2 == n**2:  
                print(x, y)  
  
pythagoreanTriples(5)
```

3 4
4 3

